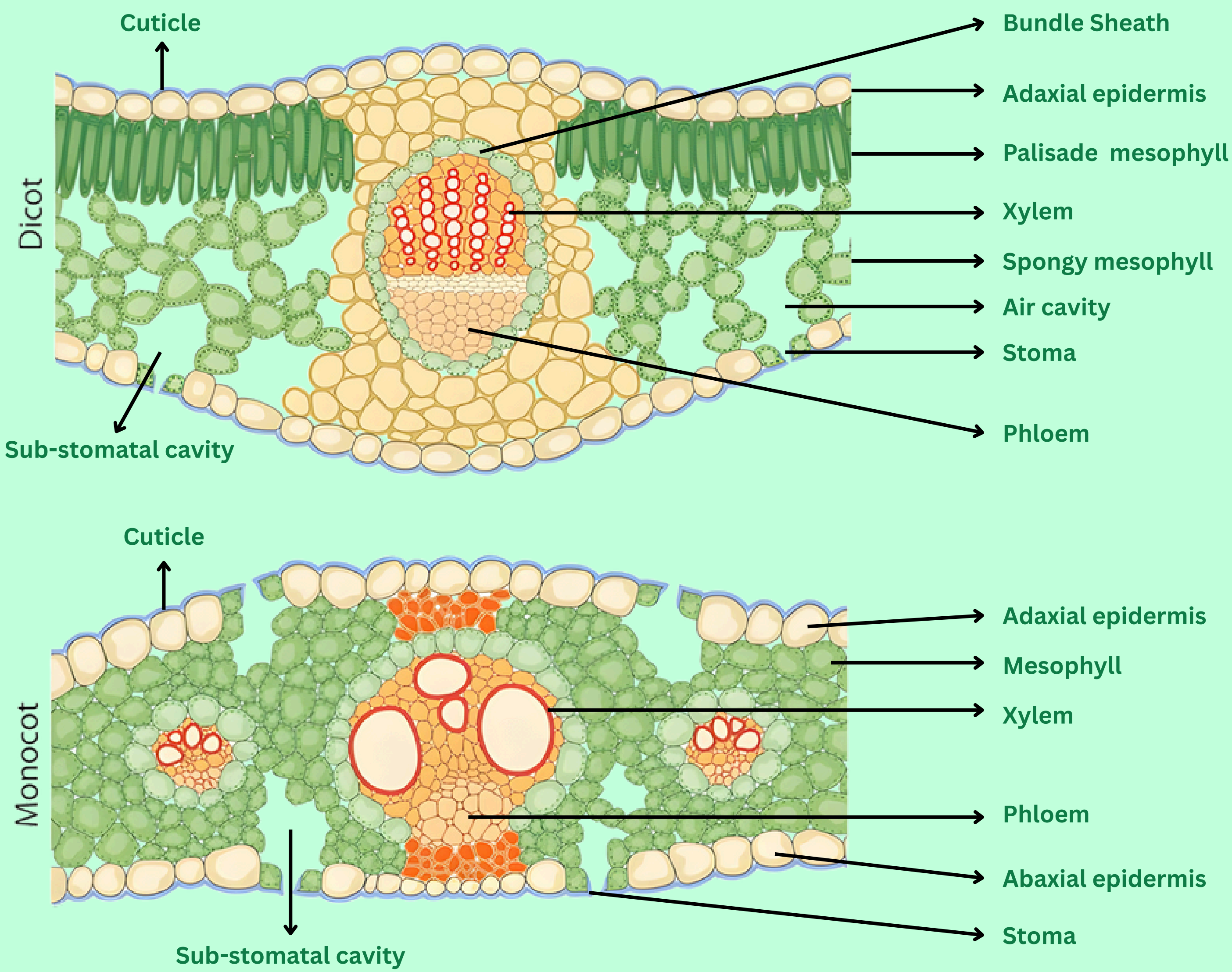
DICOT LEAF VS. MONOCOT LEAF

Feature	Dicot Leaf	Monocot Leaf
Epidermis	Dorsiventral: upper & lower epidermis	Isobilateral: both epidermal sides similar
Cuticle	Thick on upper surface	Usually uniformly thick
Stomata	More on lower surface (hypostomatic)	Equal on both surfaces (amphistomatic)
Mesophyll Differentiation	Differentiated: Palisade + spongy parenchyma	Undifferentiated mesophyll
Palisade Parenchyma	Present, 1–2 layers, main photosynthetic tissue	Usually absent
Spongy Parenchyma	Present with air spaces	Absent

Bulliform Cells	Absent	Present on upper epidermis → help in leaf rolling (water stress adaptation)
Vascular Bundles	Differ in size due to reticulate venation,	Nearly similar sized due to parallel venation
Bundle Sheath	Thin-walled parenchyma; Kranz anatomy absent (except C4 dicots)	Thick-walled; in C4 monocots shows Kranz anatomy
Xylem Location	Upper side of leaf (adaxial)	Upper side (adaxial)
Mechanical Tissue	Collenchyma in midrib and petiole	Sclerenchyma associated with bundle sheaths
Leaf Anatomy Type	Dorsiventral leaf	Isobilateral leaf

Diagram



Dicot
Cuticle
Bundle Sheath
Adaxial epidermis
Palisade mesophyll
Xylem
Spongy mesophyll
Air cavity
Stoma
Phloem
Sub-stomatal cavity
Monocot
Cuticle
Adaxial epidermis
Mesophyll
Xylem
Phloem
Abaxial epidermis
Stoma
Sub-stomatal cavity